HOTELPRO: LUXURY HOTEL MANAGEMENT & BOOKING SYSTEM

FULL STACK SOFTWARE ENGINEERING
CAPSTONE PROJECT REPORT

PROJECT DOCUMENTATION

A structured report covering the project description, real-world scenario, software requirements, system architecture, reservation workflow, database schemas, and project conclusion.

1. PROJECT DESCRIPTION

HotelPro is an enterprise-grade full-stack hospitality web platform engineered to address the operational and consumer-facing challenges of luxury hotel discoverability, room availability scheduling, reservation lifecycle management, and administrative inventory administration. Built on the MERN stack (MongoDB, Express.js, React 18, Node.js), the platform implements a decoupled, three-tier cloud-native architecture designed for high availability, security, and responsive cross-device usability.

2. REAL-WORLD SCENARIO & USE CASE NARRATIVE

In contemporary hospitality operations, boutique hotels frequently struggle with opaque property management systems that are expensive to license, difficult to maintain, and poorly integrated with modern customer booking workflows.

Scenario:

A luxury boutique resort operating 50 luxury suites across five categories (Deluxe, Super Deluxe, Suite, Family Suite, Executive) needs to manage seasonal reservations without paying high commissions to third-party travel agencies.

The Guest Experience

Travelers navigate to the resort's public web application, browse high-resolution suites through an interactive carousel, filter rooms based on desired check-in/out dates, and view transparent total prices. Once authenticated, the guest books the suite using a secure demo checkout workflow and immediately receives a unique booking confirmation reference (e.g., BK_970098). They can review active bookings and cancel reservations anytime from their dashboard.

The Operations Manager Experience

The hotel operations manager logs into a protected administrative portal authenticated via role-based access control. The dashboard provides a KPI ribbon showing total platform revenue, total bookings, active room capacity, and registered users. The manager can add new rooms, modify unit rates, and manage guest reservations across the system.

3. SOFTWARE REQUIREMENTS SPECIFICATION (SRS)

3.1 User Interfaces & Design System

- Client UI built with React 18 and Vite for fast bundle delivery and hot-reloading.
- Component styling implemented using Bootstrap 5, Material-UI (MUI), and MDB React UI Kit for consistent cross-browser design.
- Client-side routing managed by React Router v6 with custom navigation guards (ProtectedRoute.jsx, AdminRoute.jsx).

3.2 Hardware Interfaces & Deployment Topologies

- Client Deployment: Hosted on Vercel's Edge Content Delivery Network (CDN) with HTTPS termination and automatic single-page application route rewrites.
- Server Deployment: Hosted on Render Web Services running a containerized Node.js execution runtime.
- Database Cluster: Hosted on MongoDB Atlas (AWS Region ap-south-1) with TLS connection encryption and automated backup snapshots.

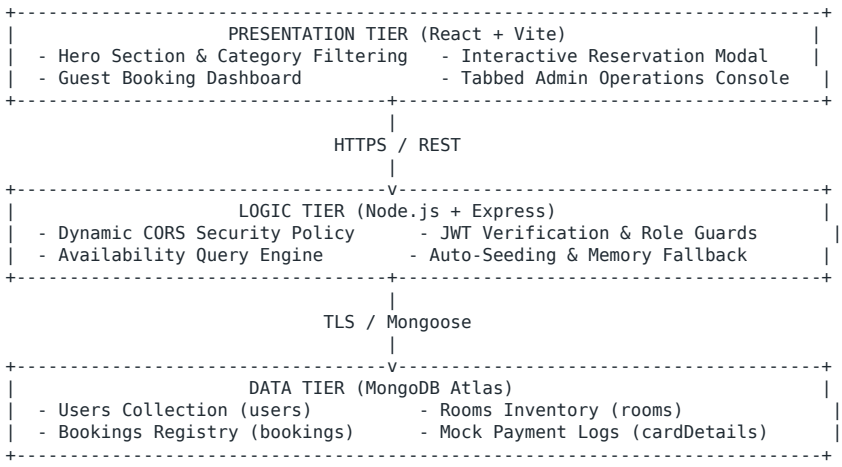
3.3 Communication Interfaces & Security Protocols

- Client-server communication operates strictly over HTTPS using RESTful JSON payloads via Axios.
- Authentication utilizes stateless JSON Web Tokens (JWT) passed via HTTP Authorization: Bearer and auth-token headers.
- User passwords are one-way hashed using bcryptjs with salt rounds set to 10.

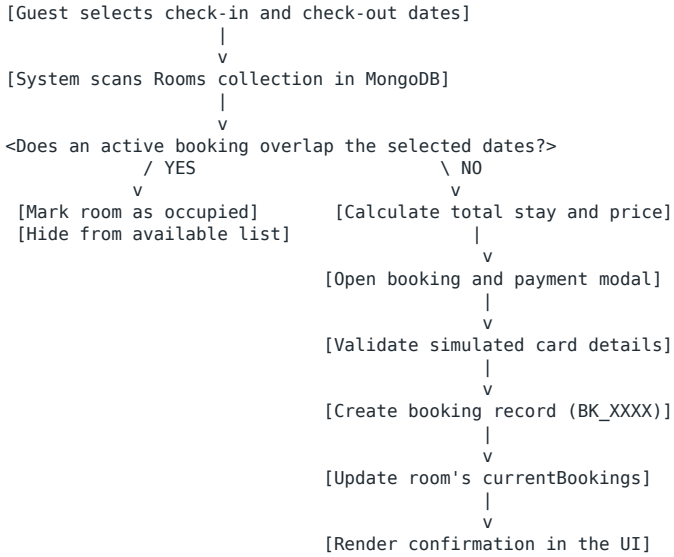
- Cross-Origin Resource Sharing (CORS) enforces dynamic origin filtering, safeguarding the backend from unauthorized cross-site requests.

4. SYSTEM ARCHITECTURE & DATA FLOW DIAGRAMS

4.1 3-Tier Layered Architecture Diagram



4.2 Functional Flowchart: Room Availability & Reservation Workflow



5. VERIFIED DATABASE SCHEMAS

The following schema fields are included as supplied in the project documentation.

User Schema

Field	Type / Constraint
user_id	String, Unique
name	String
email	String, Unique
password	Bcrypt Hash
phone	String
type	Enum: admin | customer

Room Schema

Field	Type / Constraint
room_id	String, Unique
roomNumber	String
type	String
rent	Number
count	Number
description	String
photos	Array of Strings
currentBookings	Array of Booking References

Booking Schema

Field	Type / Constraint
booking_id	String, Unique
room_id	Reference to Room
user_id	Reference to User
name	String
fromdate	String
todate	String
totaldays	Number
totalAmount	Number
status	Enum: booked | cancelled

6. CONCLUSION & PROJECT HIGHLIGHTS

The HotelPro system provides a complete, modern solution for hospitality booking and property administration. By pairing a responsive React front-end with an Express/MongoDB backend, the platform demonstrates full-stack software development best practices including clean architecture, secure authentication, dynamic data modeling, automated cloud deployments, and resilient database fallback design.